[INCHPOUND]
A-A-59358
27 August 2003

COMMERCIAL ITEM DESCRIPTION

UNDERSHIRT, MID WEIGHT, SPEAR System, Lightweight Environmental Protection (LEP), Layer-2

1. SCOPE

1.1 Scope. This Commercial Item Description covers the requirements for Layer 2- Mid Weight Undershirt fabricated in 100% Visa® Polyester Double Knit Interlock to wick moisture / perspiration away from the body. The undershirt is intended for wear by male and female military personnel of the Department of Defense as an insulation layer of the Extended Cold Weather Clothing System (ECWCS).

2. CLASSIFICATION The Undershirt shall be color Black 470 or Camouflage Green 483, as specified. The undershirt shall be of one type in the following sizes as specified:

SCHEDULE OF SIZES

Small- Regular Medium-Regular Large-Regular Large-Long
Extra Large-Regular Extra Large-Long.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center Philadelphia, Clothing and Textiles Directorate, Attn: DSCP-COCT, 700 Robbins Ave Philadelphia, PA 19111-5096.

AMSC N/A FSC 8415

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3. SALIENT CHARACTERISTICS

3.1 Design and Construction. The undershirt shall have a low bulk, fold over collar, contoured longer tail in back, long sleeves and a neck to chest zipper opening to allow ventilation. All joining seams shall be flat locked construction to reduce chaffing and improve comfort during movement.

3.2 Basic Material. The following material shall be used to provide the environmental protection, user comfort; lightweight and durability needed for wear in field, combat, and operations other than war. The basic material for the undershirt shall be 100% polyester double knit interlock, Milliken Visa® style# 696374 or product with equivalent performance characteristics. The contracting officer must approve any alternate material with equivalent performance characteristics. The fabric shall be Black 470 or Camouflage Green (CG-483). The material provides the required military appearance and functionality required by the military.

The fabric shall conform to the following requirements:

Fabric Specifications

Characteristic	Requirement	Test Method
Fiber Content	100% Visa® Polyester. Yarn Source U.S.A.	
Construction	Double Knit Interlock	
Width	60 inches minimum 62 inches maximum	
Weight per square yard Weight per linear yard	5.65 +/- .25 ounces 9.75 +/- .25 ounces	ASTM D3776
Pilling	1.0 for 30 minutes after 1 wash	ASTM D-3512, Random Tumble Method
Soil release (after 5 washes)	4.0	AATCC TM 135-3 cycles, 2, III, A,iii
Moisture transport (after 5 washes)	1.0 vs a standard of =< 5.0	AATCC TM 135-3 cycles, 2, III, A, iii
Stretch	Wales - 16.5% Courses - 92.5%	ASTM D2594, Form fit
Dimensional Stability	Wales - 2.8% max. Courses - 4.0% max.	AATCC TM 135-3 cycles, 2, III, A, iii
Air permeability test	395 cu.ft./min/sq.ft.	ASTM D737
Water Vapor Transmission Test Run under test conditions of 70º F and 50% RH for	24 hours (upright); calculations for the loss per sq. meter over 24 hour period = 330.3 grams.	ASTM E96
Colorfastness: Laundering Crocking, Dry & Wet	 4.0 minimum 4.0 minimum	 AATCC TM61-Option 2a, 3 cycles AATCC TM 8

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Perspiration, Acid	4.0 minimum	AATCC TM 15
Light, Xenon	4.0 minimum, after 80 kilojoules	AATCC TM 16E

3.3 Thread. Thread for needle and bobbin (looper) shall be commercial 100% textured polyester thread, size Tex-24 with minimum breaking strength of 2.1 lbs. conforming to type I, class 1 of A-A-52095. The thread shall be non-staining and show good colorfastness to laundering (after 3 cycles) when tested in accordance with ASTM D 204 except the specimen should be dried after each of the 3 laundering cycles and the color transfer cloth evaluation should not apply. The thread color shall match the color of basic material.

3.4 Labels. Each undershirt shall have a commercial size label, an identification label, and a care label. The size and identification labels may be combined. The identification label shall contain the item description, contract number, NSN number, fiber content information, and the contractor's name. The size label color shall be white. The inscription shall have a minimum font size of 10 points. The inscription legibility, label and label attachment shall last the expected life of the garment. If available, a basic material supplier care label and hang tag may be used, except the information must be as cited in paragraph 3.4.1.

3.4.1 Care Label. The care/instruction label for Undershirt shall read:

Machine wash warm,
Permanent press cycle.
Tumble dry medium temp.
Permanent press cycle,
Remove promptly.
Non-chlorine bleach only,
Warm iron if needed.
Do not use fabric softeners

3.4.2 Size Label. The size label shall contain, as a minimum, the following information, as applicable:

SMALL REGULAR

UNDERSHIRT, MIDWEIGHT, LAYER 2
100% POLYESTER
CONTRACT #: xxxxxx-xx-x-xxxx
NSN #: xxxx-xx-xxx-xxxx
Contractor's Name

3.4.3 Label/Tag. Each undershirt shall have a bar-code label affixed to the garment. The barcode label shall be clearly legible and readable by scanner as well as human readable. The bar coding element shall be a 13 digit national stock number (NSN). The label shall comply with the Department of Defense Logistics Applications of Automated Marking and

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Reading Symbols (DoD LOGMARS) symbology. DoD LOGMARS is a worldwide standard for government item identification. It must be located so that it is completely visible when the garment is folded and /or packaged and shall cause no damage to the item.

3.4.4 Label Placement. The labels shall be securely attached to the undershirt and positioned as follows:

> Care label and Identification/Size label shall combined and attached to center back neck (± 3/4 inch off center) and caught in collar setting seam.

3.5 Slide Fastener. The slide fastener for the front opening shall be a continuous element Polyester mono-filament, conforming to type I, style 7 of A-A-55634. Slide fastener length for the front opening is 15 inches for all sizes.

3.6 Patterns. The government shall furnish a complete set of patterns or a master pattern with grade rules, to maintain uniformity and consistency in manufacturing. The government patterns shall be used to create the contractor's working patterns. Minor modifications are permitted to accommodate manufacturing procedures, however the design and finished measurements shall be maintained.

TABLE- I. List of Pattern parts.

Material / Item	Nomenclature	Cut Parts
100%Visa® Polyester / Double Knit Interlock	Front Back Sleeve Collar Collar Tab (zipper flap) Front facing	1 1 2 1 2 2

3.7 Configuration. The following specifics are needed to provide uniform appearance; comfort and durability in use during field duties, combat, and operations other than war. End item construction and appearance shall conform to Figure 1–Undershirt.

3.8 Seaming. The seams shall be consistent, exhibit a uniform appearance and conform to the ASTM D-6193, Stitch and Seam Types listed in the Construction of Undershirt below. All material edges shall not ravel.

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TABLE – II. Constructions of Undershirt

No.	Operation	Stitch Type	Seam Type
	UNDERSHIRT		
1.	Join shoulders together.	607	FSa-1
2.	Set collar to neck with label.	301	SSa-1
3.	Join collar tab pieces together and top stitch.	301	SSe-2 (a/b)
4.	Over edge front facing.	504	EFd-1
5.	Join front facing to collar .	505	SSa-1
6.	Turn collar inside and sew to back neck and facing.	301	SSa-1
7.	Set front zipper with collar tab.	301	LSbr-3
8.	Set sleeves, close sleeves and side seams.	607	FSa-1
9.	Hem 1/2 inch ± 1/8 inch the edge of sleeves, and bottom of the undershirt.	605	EFa-2

4. REGULATORY REQUIREMENTS

4.1 Recycled, Recovered, or Environmentally Preferable Materials. Recycled, recovered, or Environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

5. PRODUCT CONFORMANCE

5.1 Product Conformance. The products provided shall meet the salient characteristic of this commercial item description, and shall conform to cited patterns, specifications, standards and quality assurance practices. The Government reserves the right to require proof of such conformance.

5.2 Quality Conformance Inspection. Sampling for inspection shall be performed in accordance with ANZI/ASQCZ1.4, as defined by contract, except where otherwise indicated.

5.3 Component and End Item Inspection. In accordance with 3.2 and 5.1, components and end items shall be tested in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable procurement documents. The Government reserves the right to inspect all components and end items to determine conformance to requirements.

5.4 End Item Visual Examination. The Undershirts shall be examined for the major defects listed below.

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EXAMINATION	DEFECT
Material and Workmanship	Component part omitted, distorted, full, tight, or twisted; any part of Undershirt caught in any unrelated stitching, the edge of any component part required to be forced out having folds of more than 1/8 inch .
	Hole, cut, tear, smash, burn, drill hole, run, thin place, dye streak, color not as specified, misweave affecting appearance or serviceability.
	Seam: puckered, distorted, pleated, wavy, twisted, irregular or open, loose or tight stitch tension, broken or missing thread or stitch, needle chew, visible mend, edge or raised stitching sewn too close to edge, resulting in damage to cloth, seam allowance not as specified, raw edge affecting appearance or serviceability.
Cleanness	Spot, stain, excessive thread ends not trimmed or removed, odor, affecting appearance or serviceability.
Shade	Shade variation within a part or between parts. Thread color not as specified.
Labels	Omitted, incorrect, illegible, not attached where specified; bar-codes omitted, not readable by scanner; human-readable interpretation (HRI) omitted or illegible; bar- code not visible on folded, packaged item; bar-code attachment causes damage to the item.
Packaging	Any Undershirt not packaged in accordance with the contract or purchase order.

5.5 Finished Dimensions. The finished Mid Weight Underwear shall conform to the measurements in inches listed in the Table-III.

Table - III. Undershirt Finished Measurements

SIZE	1/2 - CHEST (1)	1/2 - NECK (2)	BACK LENGTH (3)
SMALL	19-1/2”	7-7/8”	31-1/2”
MEDIUM	21”	8-1/4”	32”
LARGE	22-1/2”	8-5/8”	32-1/2”
X-LARGE	24-1/2”	9-1/8”	33”
LARGE-LONG	22-1/2”	8-5/8”	34-1/2”
X-LARGE-LONG	24-1/2”	9-1/8”	35”
TOLERANCE	± 1/2”	± 1/4”	+ 1”/- 3/4”

(1) Chest measurement taken from armseye to armseye with garment laid flat.
(2) Neck measurement taken along neckline folded edge to folded edge with garment laid flat.
(3) Back Length measurement taken from center back of collar seam to bottom of shirt with garment laid flat.

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Collar Height: 4-1/8 inches, ± 1/4 inch, for all sizes.

6. PACKAGING

6.1. Packaging, Packing, Marking and Palletization. Packaging, packing marking and palletization shall be in accordance with ASTM D 3951. Each pair of undershirts shall be packed individually in an commercially acceptable plastic bag and thirty (30) undershirts shall be packed in each box type CF or SF, class domestic, style RSC (variety sw with dw liner), grade 275.

6.2 Marking. Marking shall be in accordance with MIL-STD-129

7. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory unless otherwise stated in the contract or purchase order.)

7.1 Intended use: The undershirts are for wear by military personnel of the Department of Defense. They are intended for Special Operations Forces Personal Equipment Advance Requirements (SPEAR). They are used as an insulation layer of the Extended Cold Weather Clothing System in field, combat, and operations other than war.

7.2 Acceptance criteria. Sampling for inspection shall be in accordance with ANZI/ASQC Z1.4. The end items shall be examined for visual defects listed in paragraph 5.4. The lot size shall be expressed in units of undershirts. The sample unit will be one undershirt. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects. The end item shall be examined for dimensional conformance in accordance with paragraph 5.5. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of undershirts. The sample size shall be one undershirt. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0. Requirements for this acceptance criteria shall be referenced in the contract or purchase order.

7.3 Government documents.

7.3.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the

7.3.2 Acquisition Streamlining and Standardization Information System (ASSIST) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Commercial Item Descriptions (CIDS)

A-A-52095	-Thread, Polyester, Textured
A-A-55634	- Zippers (Fasteners, Slide Interlocking)

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Source of Government documents. Copies of Military and Federal documents are available from:

Standardization Documents Order Desk
Bldg 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

7.3.2 Other Government documents, drawings, and publications. The following Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those specified in the solicitation.

Code of Federal Regulations

(Applications for copies should be addressed to U.S. Government Printing Office, Superintendent of Documents, Mail stop: SSOP, Washington, DC 20402-9328.)

7.3.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issue of documents which are DOD adopted shall be those in the issue of the Acquisition Streamlining and Standardization Information System (ASSIST) cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the ASISTS are the documents cited in the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS
(For all inquires please contact the American Society for Testing and Materials, 100 Barr Harbor, West Conshohocken, PA 19428-2959). Website address http://www.astm.org

ASTM D 3776 Standard Test Method for Mass Per Unit Area (Weight) of Fabric
ASTM D 737 Standard Test Method for Air Permeability of Textile Fabrics
ASTM D 6193 Standard Test Method for Stitches and Seams
ASTM D 3512 Standard Test Method for Pilling
ASTM D 2594 Standard Test Method for Tension of Elastic Textiles
ASTM D 3951 Standard Test Method for Commercial Packaging
ASTM E 96 Standard Test Method for Water Vapor Transmission of Materials
ASTM D 204 Standard Test Method for Sewing Threads

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS
(For all inquires please contact the American Association of Textile Chemists and Colorists, P.O. Box 12215, Triangle Park, NC 27709-2215). Website address http://www.aatcc.org

AATCC TM 61 Colorfastness to Laundering, Home and Commercial: Accelerated
AATCC TM 8 Colorfastness to Crocking: AATCC Crockmeter Method

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AATCC 15 Colorfastness to Perspiration
AATCC TM16 Colorfastness to Light
AATCC TM 135 Dimensional Changes in Automatic Home Laundering of Woven and Knit Fabrics

ANSI/ASCQ Z1.4- Sampling Procedures and Tables for Inspection of Attributes (For all inquires please contact the American National Standards Institute, 25 West 43rd Street, 4th Floor, New York, NY 10036). Website address http://www.ansi.org .

7.3.4 Source of supply. Sources of suggested supply for the Visa® Polyester 1/

Milliken & Co.
920 Milliken Road
Spartenburg, NC 29303

Note: The contracting officer must approve any alternate material with equivalent performance characteristics.

1/ Visa® Performance Characteristics
Visa® is a registered trademark of Milliken & Co.
Visa® adds value by offering three key benefits:

1. **Moisture wicking** - Visa® polyester is hydrophilic. It wicks perspiration away from the skin, moving moisture to the outer surface of the garment where it evaporates.

2. **Soil release** - Skip the pretreatments. Visa® is machine washable, releasing stubborn stains and odors without pretreatments.

3. **Prevents soil redeposition** - Visa® prevents dirt and oil transfer in the wash cycle. With no soil redeposition, Visa® color stays bright and whites stay white – wash after wash. Do not use fabric softeners on Visa® polyester.

Additional benefits of Visa® polyester: Superior durability, Wrinkle resistance, Unbeatable color retention, Benefits won't wear or wash out. Apparel retains original weight, body and shape.

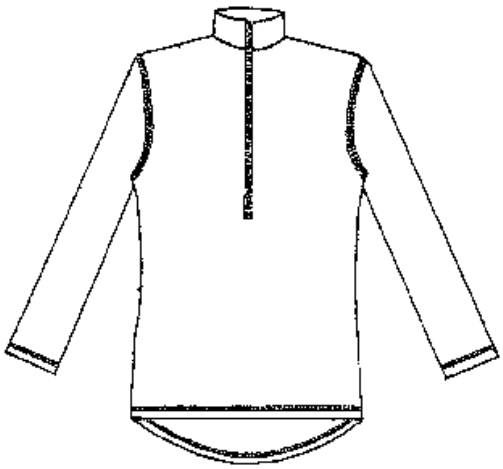
MILITARY INTERESTS:

Custodians:
Army - GL
Navy - MC

PREPARING ACTIVITY:
DLA - CT
Project 8415-0252

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FSS

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Front View



Back View

Figure 1 - Mid Weight, Undershirt

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